Aim: To develop a Flask Currency Converter Application

Q1. What is Flask and what are its key features?

Flask is a lightweight Python web framework for building web applications.

Key features include:

- Built-in development server
- URL routing
- Template engine (Jinja2)
- Request handling
- Modular design

It's classified as a micro-framework because it keeps core simple but extensible.

Q2. Explain how routing in Flask.

Routing maps URLs to functions using the `@app.route()` decorator. When a user visits a URL, Flask executes the decorated function and returns the response. Routes can include dynamic variables like `<username>`.

Q3. What are Flask templates and how does Jinja2 work?

Templates are HTML files with placeholders for dynamic content. Jinja2 enables template inheritance using `{% extends %}` and `{% block %}`, and allows inserting variables with `{{ }}` and logic with `{% %}` tags.

Q4. How does Flask handle form data from GET/POST requests?

Flask uses the request object to access form `data.request.method` checks the `HTTP` method, while `request.form['field']` retrieves `POST` data. `GET` parameters are accessed via `request.args.get()`.

Q5. How are static files (CSS, JS) managed in Flask?

Static files are stored in the static folder and linked using `url_for('static', filename='style.css')`. This generates the correct `URL` path regardless of application location.

Code (app.py):

```
from flask import Flask, render_template, request

app = Flask(__name__)

exchange_rates = {
    'USD': 1.0,
    'INR': 92.45,
    'GBP': 0.75,
    'SAR': 3.75,
    'CNY': 6.87
}
```

```
def convert_currency(amount, from_currency, to_currency):
    if from_currency == to_currency:
        return amount

    amount_in_usd = amount / exchange_rates[from_currency]
    converted_amount = amount_in_usd * exchange_rates[to_currency]

    return round(converted_amount, 2)

@app.route('/')
def hello():
    return render_template('index.html')

@app.route('/convert', methods=['POST'])
def convert():
    try:
        amount = float(request.form['amount'])
        from_currency = request.form['from_currency']
        to_currency = request.form['to_currency']

        if from_currency not in exchange_rates or to_currency not in exchange_rates:
            return render_template('index.html', error="Invalid currency selected")

        result = convert_currency(amount, from_currency, to_currency)

        currency_symbols = {
            'USD': '$',
            'INR': '₹',
            'GBP': '£',
            'SAR': '﷼',
            'CNY': '¥'
        }

        return render_template('index.html',
                               result=result,
                               amount=amount,
                               from_curr=f"{currency_symbols[from_currency]}
{from_currency}",
                               to_curr=f"{currency_symbols[to_currency]} {to_currency}")

    except ValueError:
        return render_template('index.html', error="Please enter a valid amount")
    except Exception as e:
        return render_template('index.html', error=f"An error occurred: {str(e)}")

if __name__ == '__main__':
    app.run(debug=True)
```

Code (root_layout.html):

```
<!doctype html>
<html lang="en">
    <head>
        <meta charset="UTF-8" />
        <meta name="viewport" content="width=device-width, initial-scale=1.0" />
        <title>{% block title %}WEBX EXP 7 | Flask{% endblock %}</title>
```

```
        <script src="https://cdn.tailwindcss.com"></script>

        <link
            rel="stylesheet"
            href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.1/css/all.min.css"
        />
    </head>
    <body
        class="h-screen w-screen bg-gray-100 flex justify-center items-center font-
['Segoe_UI','Calibri','Arial',sans-serif] m-0 p-0"
    >
        <div class="w-[90%] max-w-[600px]">
            {% block content %}{% endblock %}
        </div>
    </body>
</html>
```

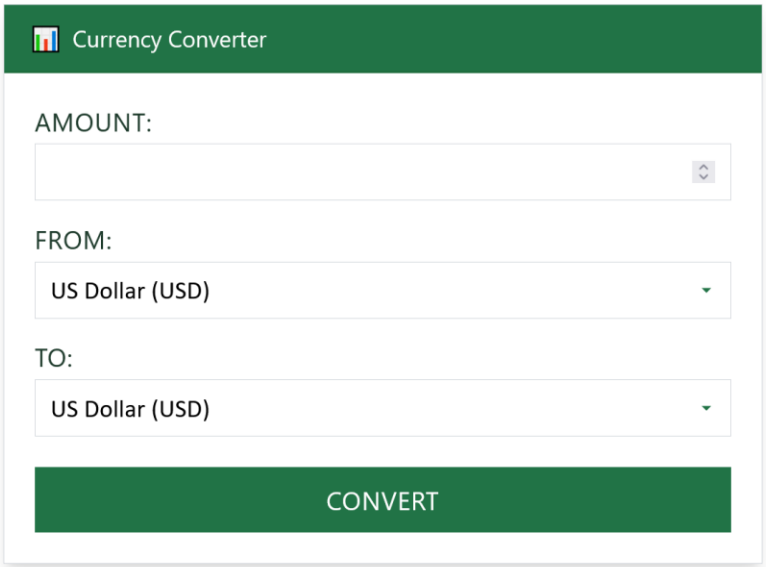
Code (index.html):

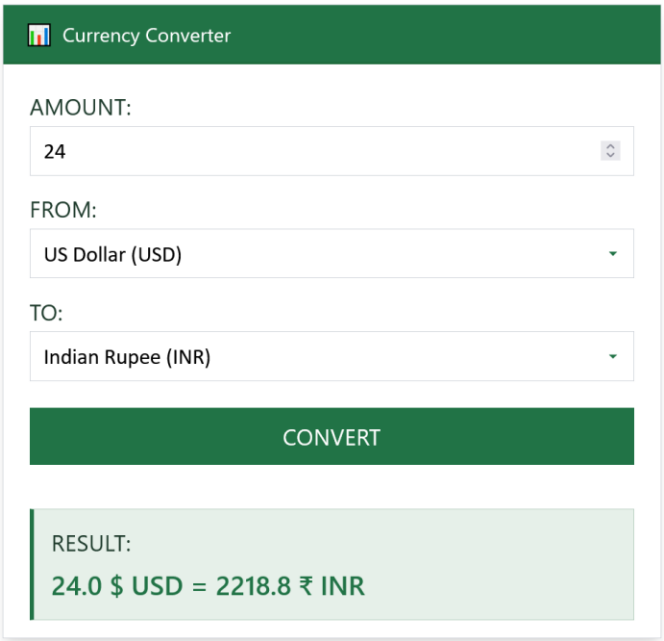
```
{% extends 'layouts/root_layout.html' %}

{% block content %}
<div class="bg-white shadow-lg border border-gray-300 overflow-hidden">

    <h1 class="bg-[#217346] text-white text-lg font-medium py-3 px-5 border-b-2 border-
[#1e6b3c] flex items-center">
        <span class="mr-2 text-xl">📊</span>
        Currency Converter
    </h1>

    <form method="POST" action="{{ url_for('convert') }}" class="p-6 bg-white">
        <div class="mb-4">
            <label class="block text-[#1e3b2c] text-xl uppercase tracking-wide mb-
1">Amount:</label>
            <input type="number" name="amount" step="0.01" min="0" required
                   value="{{ request.form.get('amount', '') }}"
                   class="w-full px-3 py-2 border border-gray-300 rounded-none text-xl
font-[Calibri] bg-white hover:border-[#217346] focus:outline-none focus:border-2
focus:border-[#217346] focus:shadow-inner">
        </div>

        <div class="mb-4">
            <label class="block text-[#1e3b2c] text-xl uppercase tracking-wide mb-
1">From:</label>
            <select name="from_currency" required
                    class="w-full px-3 py-2 border border-gray-300 rounded-none text-xl
font-[Calibri] bg-white hover:border-[#217346] focus:outline-none focus:border-2
focus:border-[#217346] appearance-none bg-no-repeat bg-right-4"
                    style="background-image: url('data:image/svg+xml;charset=UTF-8,%3csvg
xmlns=\'http://www.w3.org/2000/svg\' viewBox=\'0 0 24 24\' fill=\'%23217346\'
width=\'18px\' height=\'18px\'%3e%3cpath d=\'M7 10l5 5 5-5z\'/%3e%3c/svg%3e'); background-
position: right 10px center;">
                <option value="USD" {% if request.form.get('from_currency') == 'USD'
%}selected{% endif %}>US Dollar (USD)</option>
```

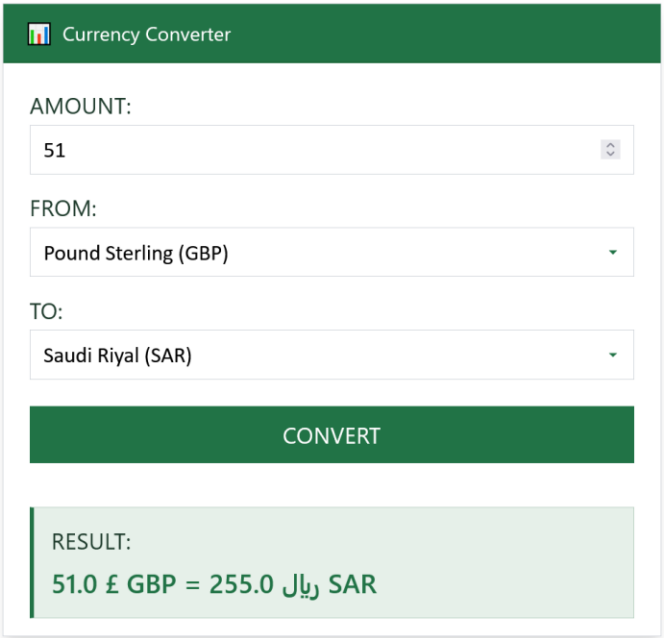
```
                <option value="INR" {% if request.form.get('from_currency') == 'INR'
%}selected{% endif %}>Indian Rupee (INR)</option>
                <option value="GBP" {% if request.form.get('from_currency') == 'GBP'
%}selected{% endif %}>Pound Sterling (GBP)</option>
                <option value="SAR" {% if request.form.get('from_currency') == 'SAR'
%}selected{% endif %}>Saudi Riyal (SAR)</option>
                <option value="CNY" {% if request.form.get('from_currency') == 'CNY'
%}selected{% endif %}>Chinese Yuan (CNY)</option>
            </select>
        </div>

        <div class="mb-6">
            <label class="block text-[#1e3b2c] text-xl uppercase tracking-wide mb-
1">To:</label>
            <select name="to_currency" required
                    class="w-full px-3 py-2 border border-gray-300 rounded-none text-xl
font-[Calibri] bg-white hover:border-[#217346] focus:outline-none focus:border-2
focus:border-[#217346] appearance-none bg-no-repeat bg-right-4"
                    style="background-image: url('data:image/svg+xml;charset=UTF-8,%3csvg
xmlns=\'http://www.w3.org/2000/svg\' viewBox=\'0 0 24 24\' fill=\'%23217346\'
width=\'18px\' height=\'18px\'%3e%3cpath d=\'M7 10l5 5 5-5z\'/%3e%3c/svg%3e'); background-
position: right 10px center;">
                <option value="USD" {% if request.form.get('to_currency') == 'USD'
%}selected{% endif %}>US Dollar (USD)</option>
                <option value="INR" {% if request.form.get('to_currency') == 'INR'
%}selected{% endif %}>Indian Rupee (INR)</option>
                <option value="GBP" {% if request.form.get('to_currency') == 'GBP'
%}selected{% endif %}>Pound Sterling (GBP)</option>
                <option value="SAR" {% if request.form.get('to_currency') == 'SAR'
%}selected{% endif %}>Saudi Riyal (SAR)</option>
                <option value="CNY" {% if request.form.get('to_currency') == 'CNY'
%}selected{% endif %}>Chinese Yuan (CNY)</option>
            </select>
        </div>

        <button type="submit"
                class="w-full bg-[#217346] text-white border-none py-3 px-5 text-xl
uppercase tracking-wide cursor-pointer transition-colors hover:bg-[#1e6b3c] active:bg-
[#185a32] rounded-none">
            Convert
        </button>
    </form>

    {% if result %}
    <div class="mx-6 my-4 p-4 bg-[#e6f0e9] border border-[#bad1c3] border-l-4 border-l-
[#217346]">
        <h3 class="text-[#1e3b2c] text-xl uppercase tracking-wide mb-2">Result:</h3>
        <p class="text-2xl text-[#217346] font-semibold break-words">
            {{ amount }} {{ from_curr }} = {{ result }} {{ to_curr }}
        </p>
    </div>
    {% endif %}

    {% if error %}
    <div class="mx-6 my-4 p-3 bg-red-50 border border-red-600 border-l-4 border-l-red-
600">
```

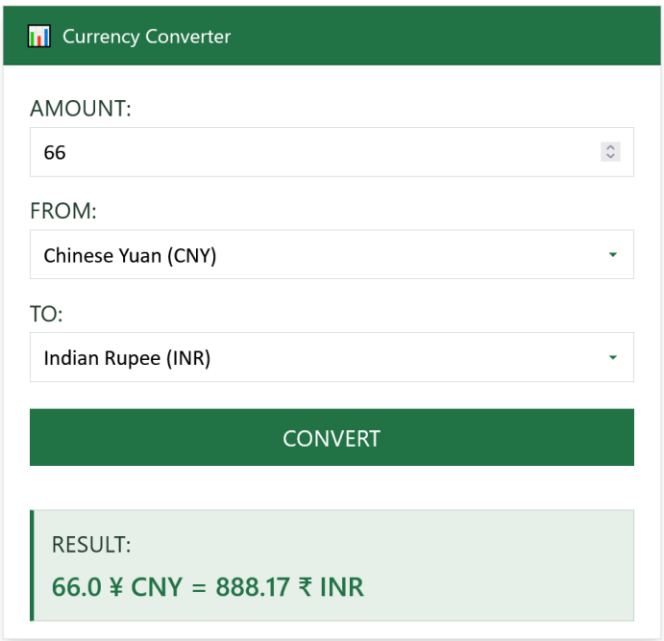
```
        <p class="text-red-600 text-xl m-0">{{ error }}</p>
    </div>
    {% endif %}
</div>
{% endblock %}
```

Output:









Conclusion:

Successfully implemented a Flask web application demonstrating core framework concepts including routing, templates, form handling, and dynamic content rendering through a functional currency converter.